

Enhancing learners' experience in e-learning based scenarios using Intelligent Tutoring Systems and Learning Analytics.

First results from a perception survey

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Abstract — E-learning students tend to get jaded and easily dropout from online courses. Enhancing the learners' experience and reducing dropout rates in these e-learning based scenarios is the main purpose of this study. This paper presents the results obtained so far and preliminary conclusions. In a first stage, the objective was to study the background and state of the art of these educational scenarios. In a second phase, identifying key reasons for dropouts, through a survey and interviews, was the aim to understand and detect motives and behavior patterns of students with dropout thoughts. Finally, developing, testing and validating a functional prototype of an Intelligent Tutoring System will allow to evaluate concepts, collect statistical information on its effectiveness, analyze and discover if course completion rates are improved.

Keywords – e-learning; dropout; MOOC; massive open online course, behavioural patterns; intelligent tutoring system; learning analytics; prototype.

I. INTRODUCTION

This research work is focused on technologies aimed at e-learning based scenarios such as Massive Open Online Courses (MOOC) or other e-learning platforms, Intelligent Tutoring Systems (ITS) and Learning Analytics (LA), and its main objective is to enhance the learners' experience and reduce dropout rates in these e-learning based scenarios. For this, it was established, as the first objective, to study the background and state of the art, mainly with the analysis, exploration and comparison among existing interactive platforms and technologies, their pros, cons and specifications.

As MOOCs and other e-learning scenarios grow in popularity, the relatively low completion rates of students has been a dominant criticism [1]. Therefore, this study aimed also to identify, in a first phase, through a survey within a student community, the key reasons for dropouts when using video lectures as a primary e-learning resource. At a second phase, further insight was obtained by interviewing teachers, counselors and platform administrators. This first research and analysis was used to detect motives and behavior patterns of students with dropout thoughts.

A very important part in a Learning Management System (LMS) is the tracking and recording of student progress with the use of learning analytics [2]. These measurements, analysis and reporting of data about learners and their contexts is essential in determining dropout patterns (in a similar way to the process of finding patterns in banking or insurance clients). Therefore, another objective of this work is to, with the usage of learning analytics, determine different stages, levels and patterns in dropout students and suggest appropriate interventions in order to prevent, in advance, these closure actions.

Based on Intelligent Tutoring Systems, and with the knowledge of abandonment patterns, computer intelligent services may be generated, in a first reaction to a dropout profile, by automatically messaging motivational sentences in an early phase; by alerting the guidance counselor and/or teacher in a middle stage, and by the usage of e-tutoring technologies (one-to-one) on an advanced stage (i.e., only within a final phase and not by student demand).

Though the work has a technological approach, its main objective is to prevent dropouts and raise completion rates within these scenarios. It also seeks that proposals follow, to the extent possible, to existing educational models and concepts. It is important to note, that the methods can vary substantially with regard to the technologies used and the target audience.

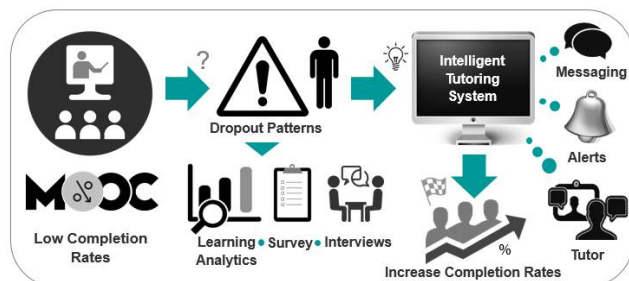


Figure 1. Motivation plan.

Figure 1 describes the motivations, steps and processes of the main plan for this research work.

This paper presents the research, analysis and work done so far on enhancing learners' experience and reducing dropout rates in e-learning based scenarios. The next section gives a brief overview of the background and state of the art on e-learning based scenarios. Section III presents the results and analysis of the survey and interviews. Section IV describes the chosen method, technologies and tools in usage on developing the prototype. Section V draws conclusions, reflects on the experience to this point and describes the future work to be done.

II. BACKGROUND AND STATE OF THE ART

The first stage in this study was to learn more about the background and state of the art of e-learning based scenarios since using, producing, and sharing Open Educational Resources (OER) has benefits for the whole educational community (i.e., individual learners, teachers, institutions, and countries) and can be seen as a systemic transformation that affects the entire educational system [3]. In this account, a thorough analysis of the literature was performed, and around 80 references were identified addressing several relevant topics:

- “Teaching, tutoring, coaching, training and mentoring”. As there's still considerable debate among the educational community on these terms, an attempt was made to clarify such terms and establish the differences and similarities among these concepts.
- “Online education background”. In order to understand the motivations, foundation and evolution of online education.
- “Learning from online lectures”. Getting in touch with a number of studies that research, seek to modify and improve education through new means and technology.
- “E-learning providers and platforms”. By identifying and classifying existing e-learning technologies, their pros, cons and specifications.
- “Highlighting text, graphic organizers and eye tracking”. Understanding general techniques that may affect cognitive processing during learning and memorization.
- “Effects of different online lecture types”. From a traditional instruction or lecture to different types of video lectures.
- “Retrieving information using speech, video and text”. Understanding methods, techniques and technologies that save, reproduce and interpret lectures.
- “Generating Open Educational Resources (OER), Collaborative Lecture Annotation System (CLAS) and Computer Supported Collaborative Learning (CSCL)”. Being familiar with technology-enabled educational resources, the extraction of important information and collaborative learning.

- “E-tutoring approach”. Understanding this unique educational method, and identifying online tutoring platforms available.
- “Parent-teacher conference/interview approach”. Recognizing the importance of ongoing home-school communication and family involvement in a school environment.
- “Learning Analytics”. By knowing the elements and purposes of understanding and optimizing learning.
- “Motivation and Dropout Studies”. By expanding knowledge on these topics through academic studies.

This first approach and study was instrumental to understand the existing context and determine, through scientific conclusions, what could be improved in actual e-learning based scenarios in order to reduce student dropouts.

III. SURVEY AND INTERVIEWS

In the next stage of this study, a total of 494 surveys were applied to college and high school students and 10 interviews were also performed with teachers, computer technicians and counselors.

The main intention with this survey and interviews was to gather information and detect motives, behaviors and patterns of possible dropout students. This research also aimed to gather professional opinions and suggestions to avoid dropping out an e-learning based course.

As shown in Figure 2, a first analysis determines that “boring contents” (47%), “demotivation” (34%), “lack of time” (28%), “bad experiences” (25%) and “can't keep up with tasks and timings” (21%) are the main reasons to quit or stop using an e-learning based system.

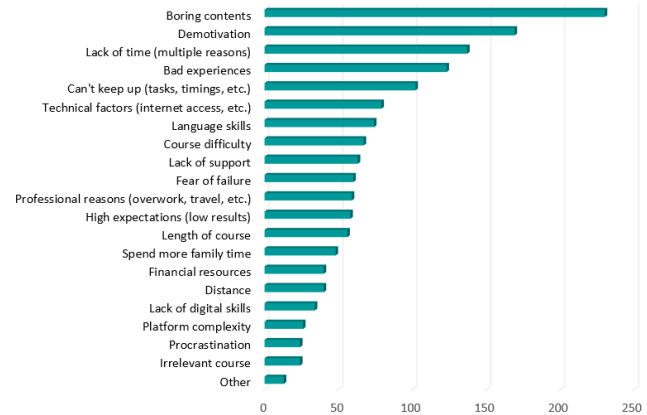


Figure 2. Main reasons to dropout.

Figure 3 reveals that, in terms of avoiding dropping out an e-learning based course, the main suggestions are: “adapting the tasks and timings” (56%), “enhance the contents” (47%), “extra support (on and offline)” (30%), “address an extracurricular course (language and/or digital skills)” (27%), “alert the teacher” (26%) and “message motivational phrases” (23%).

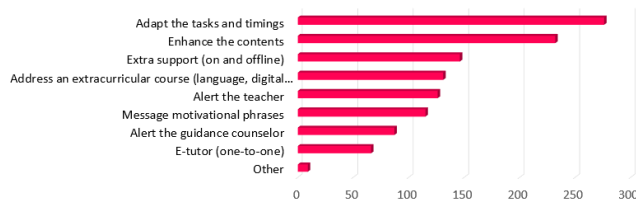


Figure 3. Suggestions to avoid dropouts.

Some research seeks to “... identify factors that predict completion of the course, for example, by conducting a correlational analysis between course completion and click stream evidence of engagement with course activities. However, that attrition happens over time. While a large proportion of students who drop out either fail to engage meaningfully in the course materials at all or drop out after the first week of participation, a significant proportion of students remain in the course longer than that but then drop out along the way. This suggests that there are students who are struggling to stay involved. Supporting the participation of these struggling students may be the first low hanging fruit for increasing the success rate of courses” [4]. Attending these students’ needs is the main purpose of this work by predicting their behaviors, patterns and avoiding dropouts.

Some studies also conclude that “... there is a correlation between the interaction of the students in using LMS and their final marks and obtained temporal behavior patterns for successful and unsuccessful students. Thus, it is possible the early detection of students who are not interacting as they are supposed to do when following a course and, consequently, to trigger alarms for both students and professors” [2].

It is also essential that students are actively involved in the learning process [5] so their continuous contribution for this or another exploration is crucial and also the initial step to start developing a functional ITS prototype agent.

IV. DEFINING A SOLUTION AND PROTOTYPING

Before coding, the first step was to define the method, possible technologies and tools to use in the development of the prototype. The chosen method was to develop a standalone agent that runs independently from the e-learning platform, which could be scalable to new features (dropout patterns) and would read logs (records of computer or software activity) from potential e-learning or MOOC frameworks.

Figure 4 portrays the prototypes’ functionalities and the interaction between agents, tasks and processes.

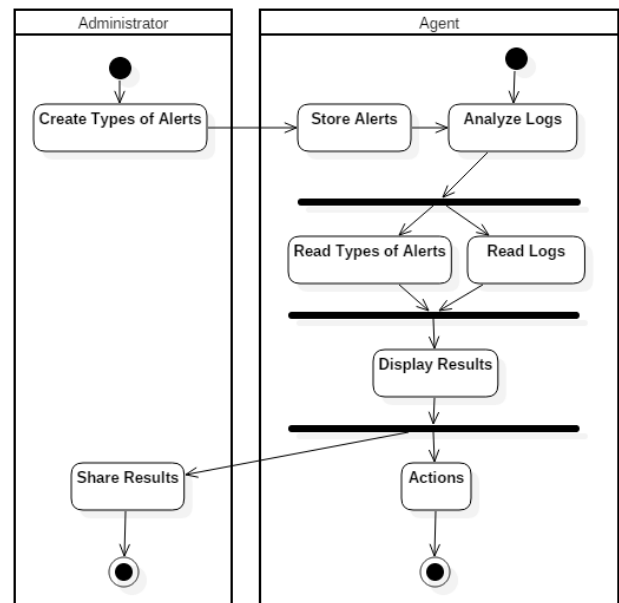


Figure 4. Agent functionality.

For a scalable and independent solution, the aim is to develop, in a multiplatform web language, an agent that would read and analyze existing logs (activities). A XML (eXtensible Markup Language) Schema represented on Figure 5 and instances were also created with a structure that defines types of dropout patterns and possible guidelines (reactions) to solve those behaviors. XML was the solution found because it can be scalable for new patterns and corresponding actions and doesn’t need a DBMS (Data Base Management System) installed.

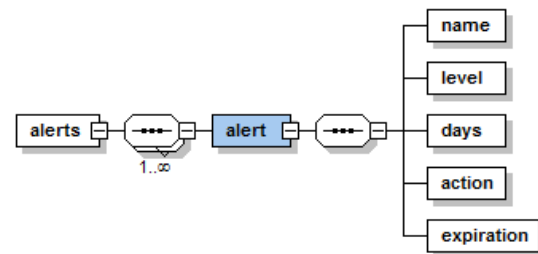


Figure 5. XML Schema Alerts.

The results (alerts) of the analysis and possible actions will be displayed to system administrators and could be shared with guidance counselors and/or teachers. These same archives will be stored in a XML format for further and future usage.

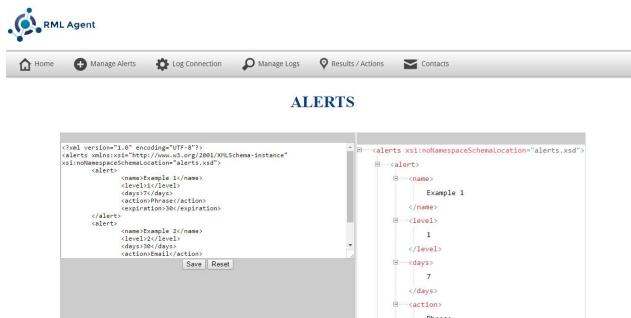


Figure 6. Prototype layout.

Currently the prototype agent (cf. Fig. 6) is in a developing stage. In its first fully operational version, it will be enriched with new features such as the “Engine Analyzer” and the “Results/Actions”.

V. CONCLUSIONS AND FUTURE WORK

The main goal of this research is to enhance the learners’ experience and reduce dropout rates in e-learning based scenarios. Learning more about the background and state of the art helped on understanding concepts, technologies and methods in this learning context. The huge participation on this survey can sustain that students are concerned with these issues and see the importance of this sort of studies. The survey and interviews also collected important data such as the main reasons to dropout and suggestions to avoid that unpleasant action. This retrieved information is being used in this early stage on the development of the prototype.

Future prototype actions, such as messaging motivational phrases in an early phase, alerting the guidance counselor and/or teacher and occasional e-tutoring support, may play an important role in avoiding dropouts. E-tutoring can be responsible for providing academic support for groups of students while they are studying a particular course through designing, delivering and evaluating e-learning activities in real-time [6].

Finally, the next step is to test and validate the prototype in real educational scenarios, with real users that meet the defined objectives and evaluate concepts within these tests. Collect statistical information on the effectiveness of current e-learning based scenarios is also an aim by comparing this information with the generated with the proposal. The results will be analyzed objectively to discover the differences between both models and attempt to measure the effectiveness of education (completion and dropout rates) before and after the implementation of the proposed solution.

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